MSc. in Computing
Practicum Approval Form

Section 1: Student Details

Project Title:	An Automated Framework for Developing and Evaluating Multimodal Retrieval-Augmented Generation Pipelines
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Chosen major:	Artificial Intelligence
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Date of Submission	21st November 2025

Section 2: About your Practicum

What is the topic of your proposed practicum? (100 words)

- The proposed practicum focuses on the design, development, and evaluation of Multimodal Retrieval-Augmented Generation (MRAG) pipelines.
- The project aims to develop 3 MRAG pipelines using Multimodal Retrieval and Multimodal LLMs, applying three different fusion techniques: Early, Late, and Hybrid Fusion
- A large, publicly available multimodal dataset will be used to develop, evaluate, and compare pipeline performance across multiple evaluation dimensions.
- The practicum will culminate in the development of a no-code/low-code platform that automates MRAG pipeline construction, testing, and selection, enabling users or developers to easily generate the most suitable pipeline for their dataset.

Please provide details of the papers you have read on this topic (details of 5 papers expected).

- We are focusing on research papers that cover the following areas:
    a. Implementation strategies for Multimodal Retrieval-Augmented Generation
    b. Evaluation methodologies for MRAG pipelines
    c. Data fusion techniques used in multimodal systems

1. **UniRaG: Unification, Retrieval, and Generation for Multimodal Question Answering With Pre-Trained Language Models**

   What is the research about: This paper is about UniRaG, a new system that helps answer questions using information from both text and images by combining them into a single approach.

   The problem they are solving: Other systems lose important information when combining text and images, or need complicated and slow methods that don't work well for all types of questions.

   Solution: UniRaG turns everything (even pictures and tables) into text, finds the most relevant facts or passages, and then uses a smart language model to write an answer.

   Key Findings: This new method works much better than older ones, giving higher accuracy when tested on a well-known dataset for questions that use both text and images.

   Methodology: The system has three steps: convert everything to text, search for the best pieces of information for the question, and generate an answer using a powerful language model.

   Special remarks: Additional tests show that using better tools to describe images, and a special kind of training, make the system even more accurate and reliable.

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2. **MuRAG:Multimodal Retrieval-Augmented Generator for Open Question Answering over Images and Text**

   What is the research paper about: This paper introduces MuRAG, a model that combines both images and text from external sources to help answer open-ended questions more accurately.

   The problem they are solving: Most current models can only retrieve and use information from text, missing out on key details present in images, which limits their usefulness for some questions.

Solution: MuRAG uses a large memory of both images and text, retrieving the most relevant content in both forms, to produce better answers than text-only models.

Key Findings: MuRAG achieves much better accuracy—outperforming previous models by 10–20%—on standard benchmarks requiring understanding of both text and visual information.

Methodology: The model is pre-trained with both image-text and only-text data using special training methods, and is tested on datasets that require mixed understanding and reasoning.

Special remarks: By showing that multimodal retrieval improves results over text-only methods, MuRAG sets a new standard for answering visual and textual questions together.

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3. **Multimodal Retrieval Augmented Generation Evaluation Benchmark**

What is the research paper about: This paper presents a new tool and benchmark for testing how well MRAG models can answer questions using both images and text.

The problem they are solving: There is no common or standard way to measure how good MRAG models are—making it hard for developers and regulators to judge their quality or robustness.

Solution: The authors built a platform with strong tests and specific metrics for MRAGs, including noise robustness, negative rejection (ignoring wrong info), integration of retrieved info, and handling "what if" scenarios.

Key Findings: Their evaluation platform reliably tells apart strong and weak MRAG models and shows exactly where models need improvement before real-world use.

Methodology: They use a multimodal feature extractor, custom test generation, and a web app to automatically run and compare tests on different MRAG systems, with results for both developers and organizations.

Special remarks: This platform helps both AI builders and regulators create better, safer MRAG models by providing clear benchmarks that encourage further improvements.

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4. **MRAMG-Bench: AComprehensive Benchmark for Advancing Multimodal Retrieval-Augmented Multimodal Generation**

What is the research paper about: This paper introduces MRAMG-Bench, a benchmark and dataset for testing and comparing how well MRAG models generate answers with both text and images.

The problem they are solving: Existing MRAG evaluation methods mostly focus on text-only results or lack strong datasets and metrics for mixed text-image answers, which limits progress on truly multimodal systems.

Solution: MRAMG-Bench offers a large, carefully designed set of multimodal documents and questions across diverse domains, along with new metrics to fairly judge how well models integrate and present both images and text in their output.

Key Findings: Using MRAMG-Bench, the authors show that current models—including popular LLMs and MLLMs—struggle with some multimodal tasks, especially handling images and text together, highlighting areas for future improvement.

Methodology: They build and use a multi-domain dataset, run tests across many leading models, and propose both statistical and language-model-based evaluation metrics to measure image and text accuracy, relevance, and integration.

Special remarks: The benchmark makes it much easier to track MRAG progress, and the analysis suggests that model design must improve for challenging, real-world mixed-media questions.

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5. **Combine early and late fusion together: An hybrid fusion framework for image-text matching**

What is the research paper about: This paper introduces a new 'hybrid fusion' method that mixes both early and late data fusion strategies for matching images and texts.

The problem they are solving: Standard models only use either early or late fusion methods, each with major drawbacks—late fusion is slow and loses information, while early fusion misses fine details.

Solution: They designed a framework called CELFT, which applies early fusion to raw image and text features, and then combines it with late fusion in a flexible, universal structure.

Key Findings: CELFT improved matching accuracy and trained faster than previous methods when tested on common image-text datasets (MSCOCO and Flickr30K).

Methodology: The system first combines visual regions and text features early on, then uses standard deep learning models to further fuse them later, and compares results with strong baselines.

Special remarks: The model works well for text-to-image and image-to-text retrieval, and its fusion approach can be added easily to existing systems to make them more powerful.

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6. **Multi-Modal Fusion for Objective Assessment of Cognitive Workload: A Review**

What is the research paper about: This paper reviews and compares methods for combining data from multiple sources (like sensors) to better assess cognitive workload in humans, highlighting fusion strategies.

The problem they are solving: Systems using only one kind of data (like EEG or heart rate) often miss important signals and can fail in real-world (noisy, incomplete) scenarios.

Solution: The paper surveys different fusion methods—feature-level, decision-level, hybrid—and explains how mixing data improves reliability, accuracy, and robustness for workload estimation.

Key Findings: While fusion improves results in some situations, many studies show only small benefits over single-modality methods; decision-level fusion (especially combining EEG and fNIRs) is most successful.

Methodology: A systematic literature review analyzed 34 papers, mapped fusion approaches against sensor combinations, and summarized which strategies worked best in different tasks.

Special remarks: The review explains why multi-modal fusion has untapped potential and recommends design choices for future fusion systems, especially for adaptive automation.

How does your proposal relate to existing work on this topic described in these papers? (200 words)

My proposal builds on recent advances in Multimodal Retrieval-Augmented Generation (MRAG) systems and the comprehensive research reviewed. Papers such as UniRaG and MuRAG introduce robust approaches for integrating retrieval and generation across text and images, but users still face complexity in selecting the best fusion and retrieval strategies for their unique datasets. Evaluation-focused works like MRAG-Benchmark and MRAMG-Bench expose the lack of standardized, fine-grained assessments for MRAG pipelines, highlighting the need for accessible tools that promote objective comparison across fusion methods and data types. The reviewed literature on fusion strategies demonstrates the nuanced trade-offs between early, late, and hybrid fusion models, which affect performance differently depending on the application context.

My proposal directly extends these works by developing a no-code/low-code platform that not only allows users (both technical and non-technical) to easily experiment with, evaluate, and select optimal MRAG pipelines but also incorporates research-backed evaluation frameworks and supports a wide range of fusion/retrieval permutations. By automating benchmarking—leveraging the metrics and datasets proposed in leading papers—my platform aims to democratize access to MRAG technology, reduce the trial-and-error burden, and empower users to make data-driven decisions that were previously only accessible to specialists, thus broadening the real-world impact of recent MRAG research.

What are the research questions that you will attempt to answer? (200 words)

Our research aims to address several key questions at the intersection of Multimodal Retrieval-Augmented Generation (MRAG), fusion strategies, and evaluation frameworks. First, we will investigate how different fusion approaches—early, late, and hybrid—impact the effectiveness, accuracy, and efficiency of MRAG pipelines across varied datasets and modalities, going beyond current work that often explores only single strategies. Building upon recent advances such as UniRaG and MuRAG, we will examine which pipeline configurations best handle multimodal input (text, images, tables) and retrieval sources for real-world tasks. Another core question is what objective benchmarks, evaluation metrics, and experimental frameworks are most appropriate for measuring MRAG pipeline performance, as highlighted by gaps in works like MRAG-Bench and MRAMG-Bench. we will explore how these can be integrated and automated to facilitate robust, user-friendly comparison.

Finally, our research will address how a no-code/low-code platform can democratize access for non-technical users—enabling them to identify and deploy the most suitable MRAG pipeline for their dataset without deep technical expertise. By answering these questions, I seek to advance both the scientific understanding and practical accessibility of MRAG systems, fostering greater adoption and innovation in multimodal AI.

How will you explore these questions? (Please address the following points. Note that three or four sentences on each will suffice.)

1. What software and programming environment will you use?

Programming Language: Python	AI Framework: Langchain	Large Language Model: Llama
Vector Database: ChromaDB	Hugging Face	Backend Framework: Flask
Vercel / AWS	Groq AI	Embedding Models
Nextjs + TailwindCSS	MERN	Firebase

2. What coding/development will you do?

- 3 Multimodal Retrieval Augmented Generation Pipelines (Multimodal Retrieval + Multimodal LLM) using 3 different fusion techniques
    a. Early Fusion
    b. Late Fusion
    c. Hybrid Fusion
- No Code / Low Code platform for development of custom Multimodal RAG pipeline.
    a. User/Developer can upload dataset
    b. Select option to automate or manually customize pipeline
    c. Develop and evaluate the pipeline

d. The user / developer can finally get the code to the MRAG pipeline and evaluation results.

3. What data will be used for your investigations?

- **Research Paper:** Benchmarking Retrieval-Augmented Multimodal Generation for Document Question Answering
  Dataset: MMDocRAG is a dataset for multimodal retrieval-augmented generation on long documents (with text, tables, charts, images).
  Dataset link: https://huggingface.co/datasets/MMDocIR/MMDocRAG

- **Research Paper:** M3DocRAG: Multi-modal Retrieval is What You Need for Multi-page Multi-document Understanding
  Dataset: Designed for multi-page and multi-document multimodal understanding. Includes a new benchmark: M3DocVQA (3,000+ PDF docs, 40,000+ pages)
  Dataset Link: https://github.com/bloomberg/m3docrag

4. Is this data currently available, if not, where will it come from?

- The data is currently available. We are still exploring multiple datasets that suit our research and development needs.

5. What experiments do you expect to run?

- We aim to test different MRAG pipelines (early, late, and hybrid fusion) on standard multimodal datasets using various combinations of retrieval and fusion strategies.
- We will compare the performance, accuracy, and speed of each pipeline and also develop a No Code / Low Code platform for quick and easy development and evaluation of MRAG pipelines.

6. What output do you expect to gather?

- We expect to collect results showing how well each MRAG pipeline performs (e.g., accuracy, response quality, speed) for the given dataset and tasks.
- The platform will also provide ready-to-use code for the best-performing pipeline chosen by the user/developer.

7. How will the results be evaluated?

Metric	**What it Measures**
Recall@k, Precision@k	Retrieval effectiveness
MRR, NDCG	Ranking quality in retrieval
EM, F1, BLEU, ROUGE	Generation accuracy
Faithfulness/Hallucination	Context grounding, factuality
Context Utilization, MM-F	Multimodal interaction, grounding
Human Evaluation	Real-world utility, multimodal use